1. Clinical Question & Core Hypothesis

Clinical Question

TNBC tumors often present with dense lymphocytic infiltrates, yet high TIL density does not uniformly translate into improved survival; the Ontologist's tiers point to a competing immunosuppressive TAM axis (CD163) and tumor-derived circulating miRNA (miR-6855-5p) as candidate modulators that could blunt TIL efficacy. 2) Tier 1 paths used: I built on the CD163/TAM path because it directly explains local immune suppression, and the miR-6855-5p/EV path because secreted miRNAs provide a plausible tumor-to-immune-cell reprogramming mechanism; I referenced the HbA1c/metabolic path for context but did not include it as a core anchor to preserve focus. 3) Tier 2 evidence supporting the claim: Hedgehog/CCND1 and proliferation programs can increase tumor intrinsic resistance to immune control (providing a proliferative competitor to immune elimination); LINE■1 methylation and systemic metabolic markers (HbA1c) support a metabolic/epigenetic milieu that may cooperate with TAMs to impair anti-tumor immunity. 4) Endpoint choice: Overall survival (OS) was chosen because it is the most robust, well-annotated clinical endpoint in TCGA-BRCA and aligns with the Ontologist's path linking miRNAs to OS. 5) Deliberate exclusions: I excluded TLS-like aggregates and TSR from the core hypothesis to avoid conflating distinct immune-architectural readouts; I relegated systemic metabolic marker HbA1c and pathway nodes (CCND1, Hedgehog, LINE-1) to secondary_modifiers to keep the mechanistic question sharp and testable with one imaging feature and up to two molecular anchors?

Working Hypothesis

In TNBC, high TIL density alone does not guarantee improved overall survival unless the tumor microenvironment lacks a dominant TAM-immunosuppressive axis and tumor-derived extracellular miRNA cargo. Specifically, TNBC tumors that are TIL-high but also CD163-high and miR-6855-5p-high will show worse OS than TIL-high, CD163-low, miR-6855-5p-low tumors. Mechanistically, CD163+ tumor-associated macrophages secrete immunosuppressive cytokines (eg IL-10, TGF-β) and create a milieu that dampens T cell cytotoxicity and antigen presentation. Concurrently, tumor-derived EVs carrying miR-6855-5p are taken up by CD8+ T cells and dendritic cells, downregulating components of IFN-γ signaling, MHC class II and co-stimulatory pathways, and cytotoxic effector programs. The net result is a decoupling of TIL presence from effective anti-tumor immunity in the subset with high CD163 and high miR-6855-5p, explaining inferior OS despite abundant TILs. Testable predictions: (i) in TNBC cohorts, the interaction effect of TIL density with CD163 expression and miR-6855-5p EV levels better predicts OS than TIL density alone; (ii) tumors with high CD163 exhibit T cell and antigen presentation gene signatures that are suppressed relative to CD163-low tumors; (iii) in vitro, EVs enriched for miR-6855-5p suppress T cell cytotoxicity and DC antigen-presentation markers, reversible by anti-miR-6855-5p inhibition; (iv) in preclinical models, targeting TAMs or inhibiting miR-6855-5p function restores TIL activity and improves survival.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CD163, miR-6855-5p
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD163: CD163/TAM path because it directly explains local immune suppression, and the miR-6855-5p/EV path because secreted miRNAs provide a plausible tumor-to-immune-cell reprogramming mechanism; I referenced the HbA1c/metabolic path for context but did not include it as a core anchor to preserve focus.

miR-6855-5p: miR-6855-5p/EV path because secreted miRNAs provide a plausible tumor-to-immune-cell reprogramming mechanism; I referenced the HbA1c/metabolic path for context but did not include it as a core anchor to preserve focus.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	HbA1c	Supporting modifier from knowledge graph
Secondary	CCND1	Supporting modifier from knowledge graph
Secondary	Hedgehog_pathway	Supporting modifier from knowledge graph
Secondary	LINE-1_methylation	Supporting modifier from knowledge graph
Secondary	MDA-MB-231 (model)	Supporting modifier from knowledge graph
Secondary	Extracellular vesicles (EVs)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective cohort analyses with multivariate Cox models testing interactions among TIL density, CD163, and miR-6855-5p EV burden on OS;

Prediction 2: single-cell or bulk transcriptomics to compare immune signatures across CD163 strata;

Prediction 3: isolation and characterization of tumor-derived EVs, cargo profiling, and in vitro assays with human T cells/DCs to assess cytotoxicity, antigen presentation, and cytokine production;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- HbA1c: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CCND1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.